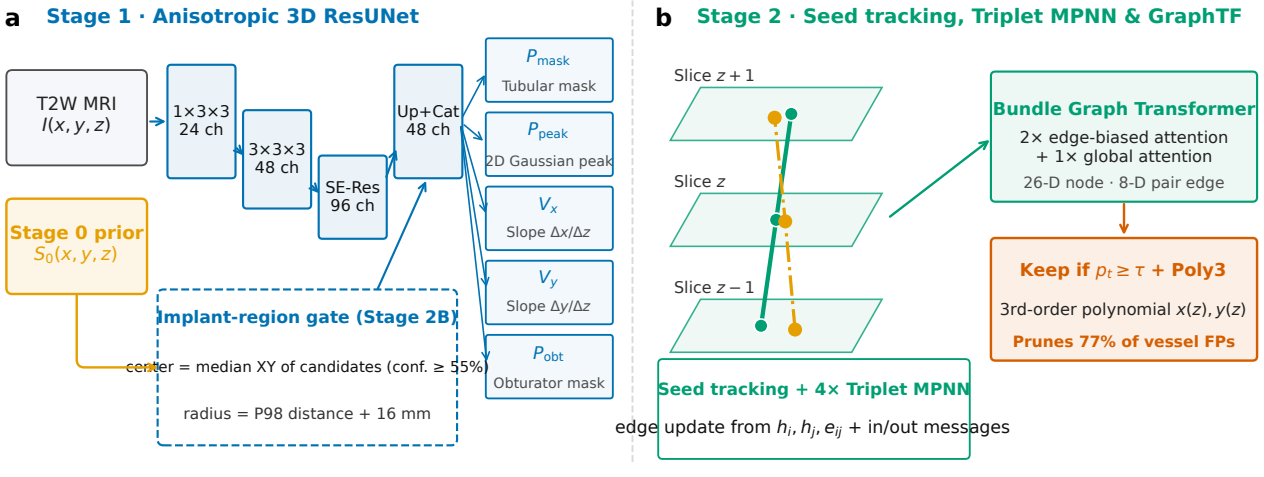
a
Stage 1 · Anisotropic 3D ResUNet
T2W MRI
$I(x, y, z)$
1×3×3
24 ch
3×3×3
48 ch
SE-Res
96 ch
Up+Cat
48 ch
$P_{mask}$
Tubular mask
$P_{peak}$
2D Gaussian peak
$V_x$
Slope $\Delta x/\Delta z$
$V_y$
Slope $\Delta y/\Delta z$
$P_{obt}$
Obturator mask
Stage 0 prior
$S_0(x, y, z)$
Implant-region gate (Stage 2B)
center = median XY of candidates (conf. ≥ 55%)
radius = P98 distance + 16 mm
b
Stage 2 · Seed tracking, Triplet MPNN & GraphTF
Slice $z+1$
Slice $z$
Slice $z-1$
Seed tracking + 4× Triplet MPNN
edge update from $h_i, h_j, e_{ij}$ + in/out messages
Bundle Graph Transformer
2× edge-biased attention
+ 1× global attention
26-D node · 8-D pair edge
Keep if $p_t \geq \tau$ + Poly3
3rd-order polynomial $x(z), y(z)$
Prunes 77% of vessel FPs